



AI Meets Aesthetics:

**Maintaining order and complexity
from abstract patterns to real-world visuals
for aesthetics research**

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| master's thesis student

| daily supervisor + promotor

| daily supervisor

| co-promotor

AI Meets Aesthetics



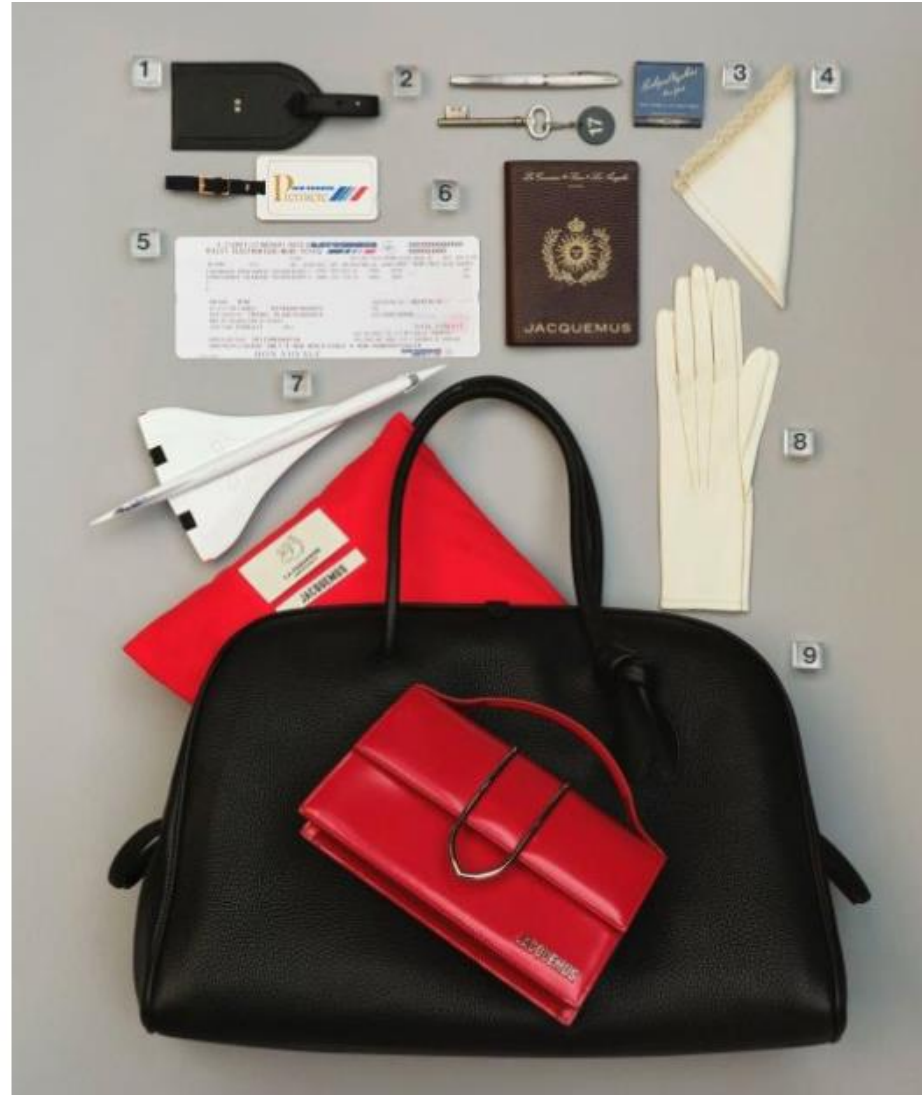
- Context overview
- Literature review
- Proposed methodology
- Initial results
- Next steps

Context Overview

Things Organized Neatly



Things Organized Neatly

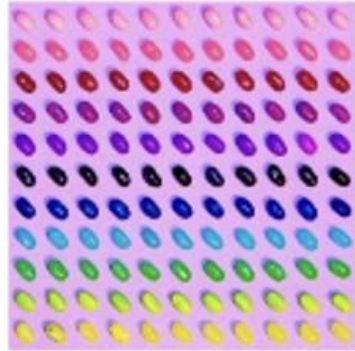


Things Organized Neatly



Context Overview

Aesthetic Preferences



What factors play a role in humans' appreciation of images?

Context Overview

Order

Order



structure and **organization**
of elements

Order and Complexity

Order



structure and **organization**
of elements

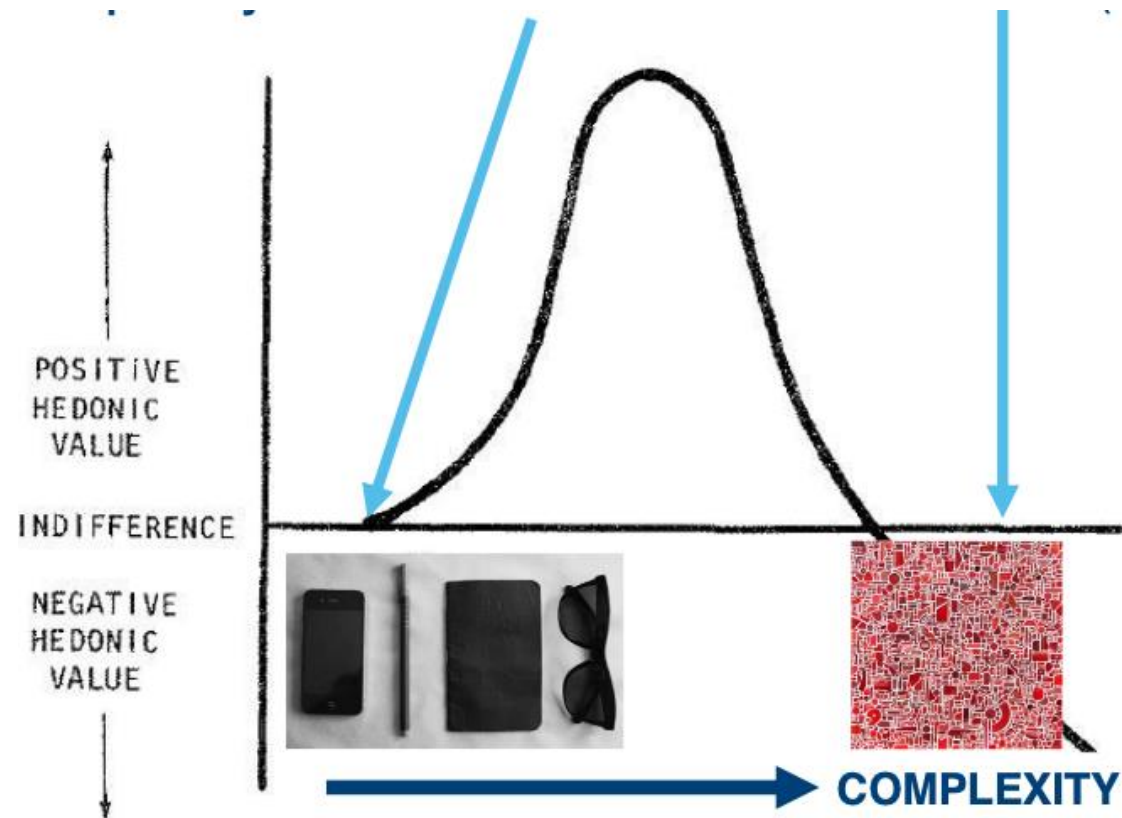
Complexity



quantity and **diversity**
of elements

Order and Complexity

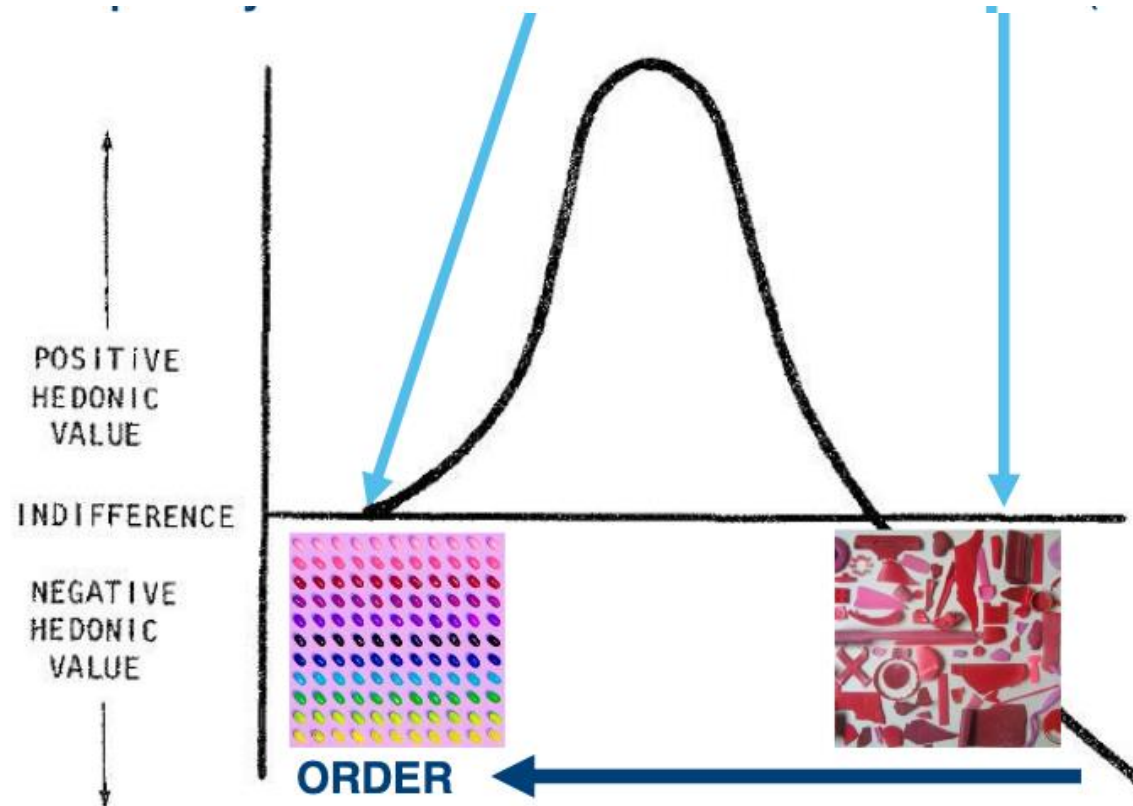
Complexity without order produces **confusion**;



(Arnheim, 1966)

Order and Complexity

Complexity without order produces **confusion**; order without complexity causes **boredom**

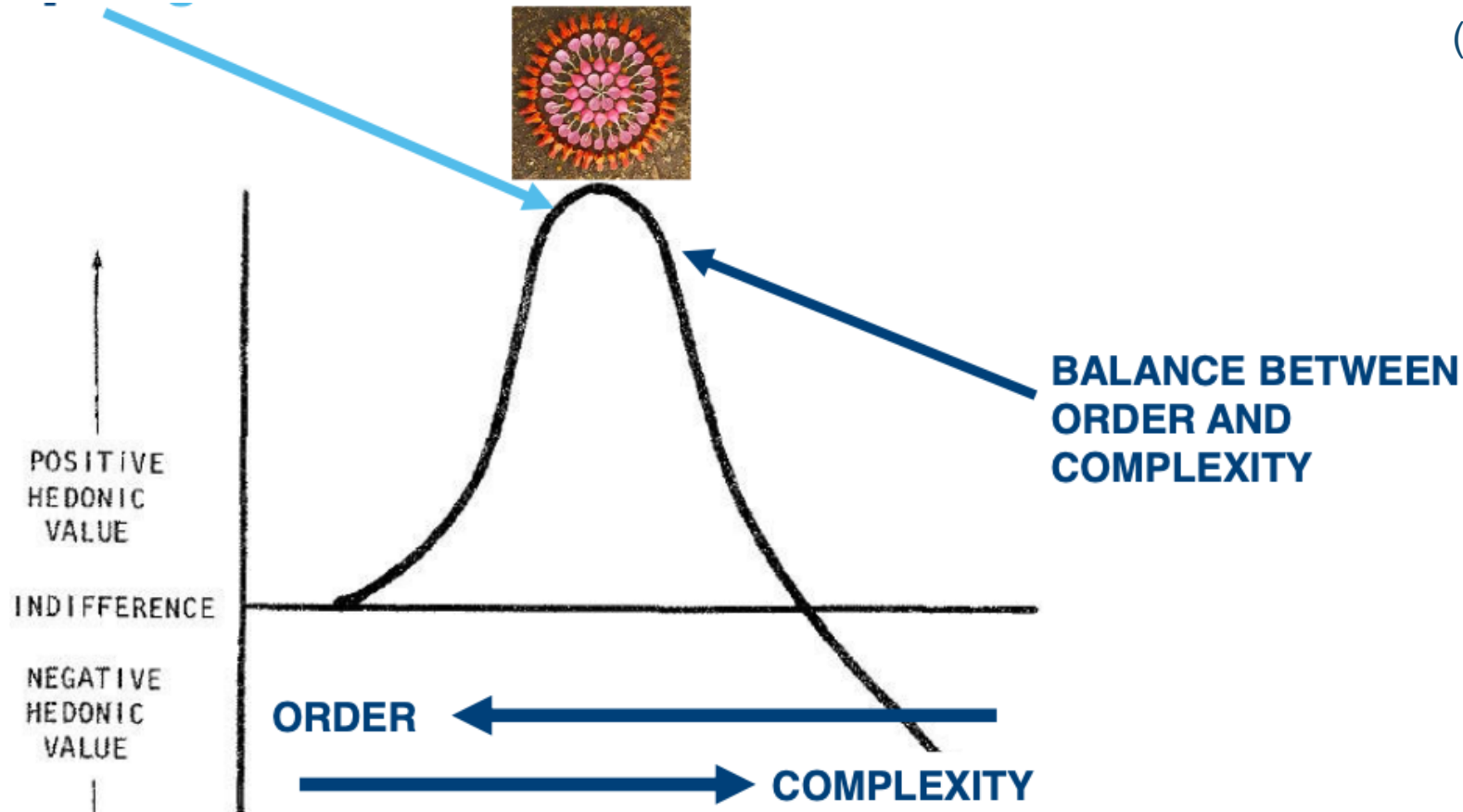


(Arnheim, 1966)

Order and Complexity

... **delight** lies somewhere between boredom and confusion

(Gombrich, 1984/1992)



Order and Complexity

However, earlier research on **order and complexity**:

- Studied them **separately**
- Studied together but **cannot disentangle the two** (not **parametrically controlled**)

Order and Complexity

However, earlier research on **order and complexity**:

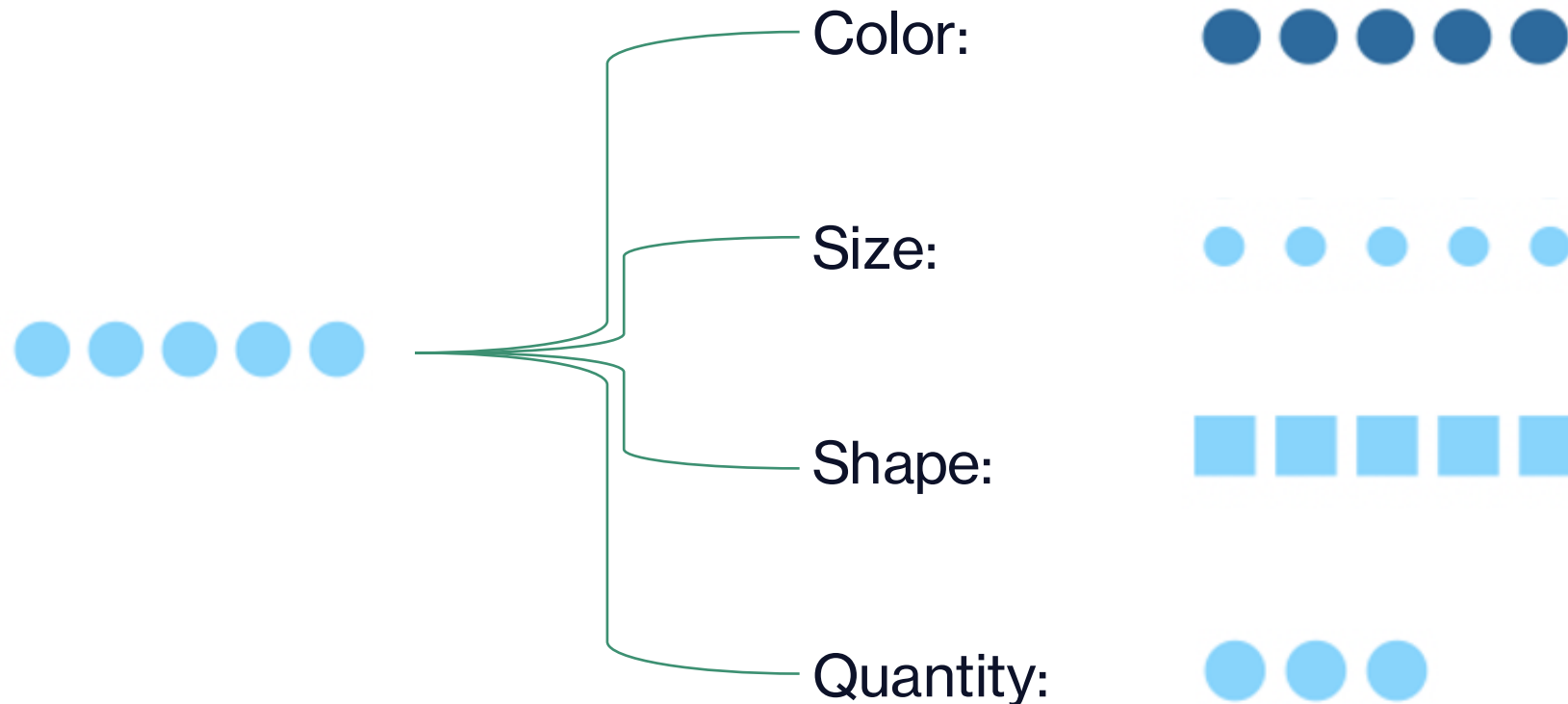
- Studied them **separately**
- Studied together but **cannot disentangle the two** (not **parametrically controlled**)

What **parametrically controlled** means exactly?

Ability to manipulate the order / complexity quantitatively based on specific features

Parametric control

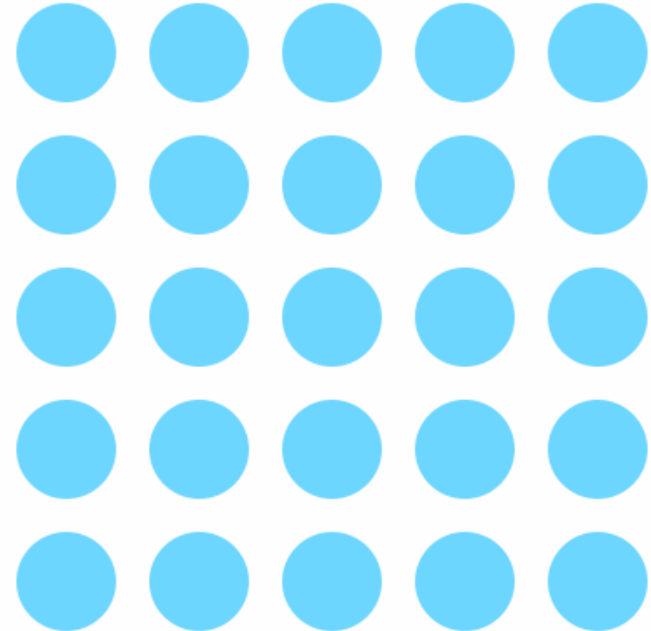
Ability to manipulate the order / complexity **quantitatively** based **on specific features**,
e.g.: **color, size, shape, quantity, distance**:



Parametric control & OCTA Toolbox

With **parametric control**, what can we do?

Adjust features quantitatively to control **order** and **complexity** level



Order: ★★★★★

Complexity: ★☆☆☆☆

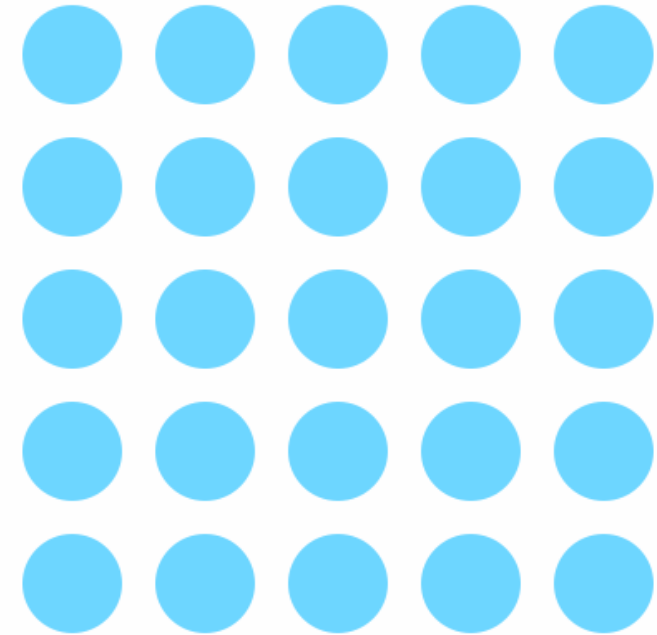
Parametric control & OCTA Toolbox

This parametric control is introduced through:

OCTA Toolbox (Van Geert, Wagemans, 2023)

It has since enabled research on:

**humans' preference towards aesthetic based on
order and complexity combination**



Order: ★★★★★

Complexity: ★☆☆☆☆

Problem Statement



Problem Statement

Revisiting the pebbles from *Neatly Organized Things* earlier:

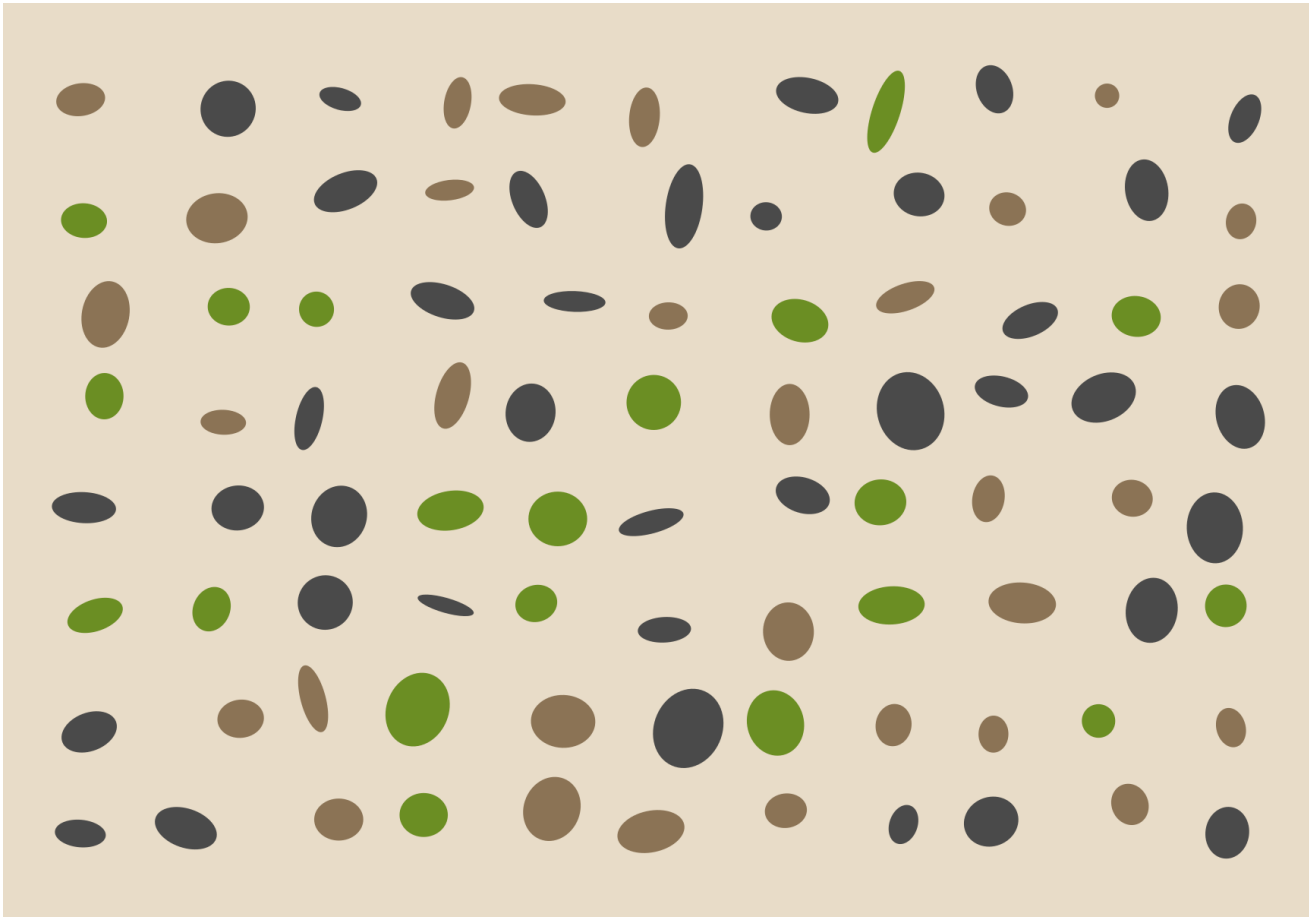


This is a **real-world photograph** of a neatly organized collection of pebbles.

- It is **realistic**, but
- **not parametrically controlled**; provides **very limited flexibility** for order vs complexity study

Problem Statement

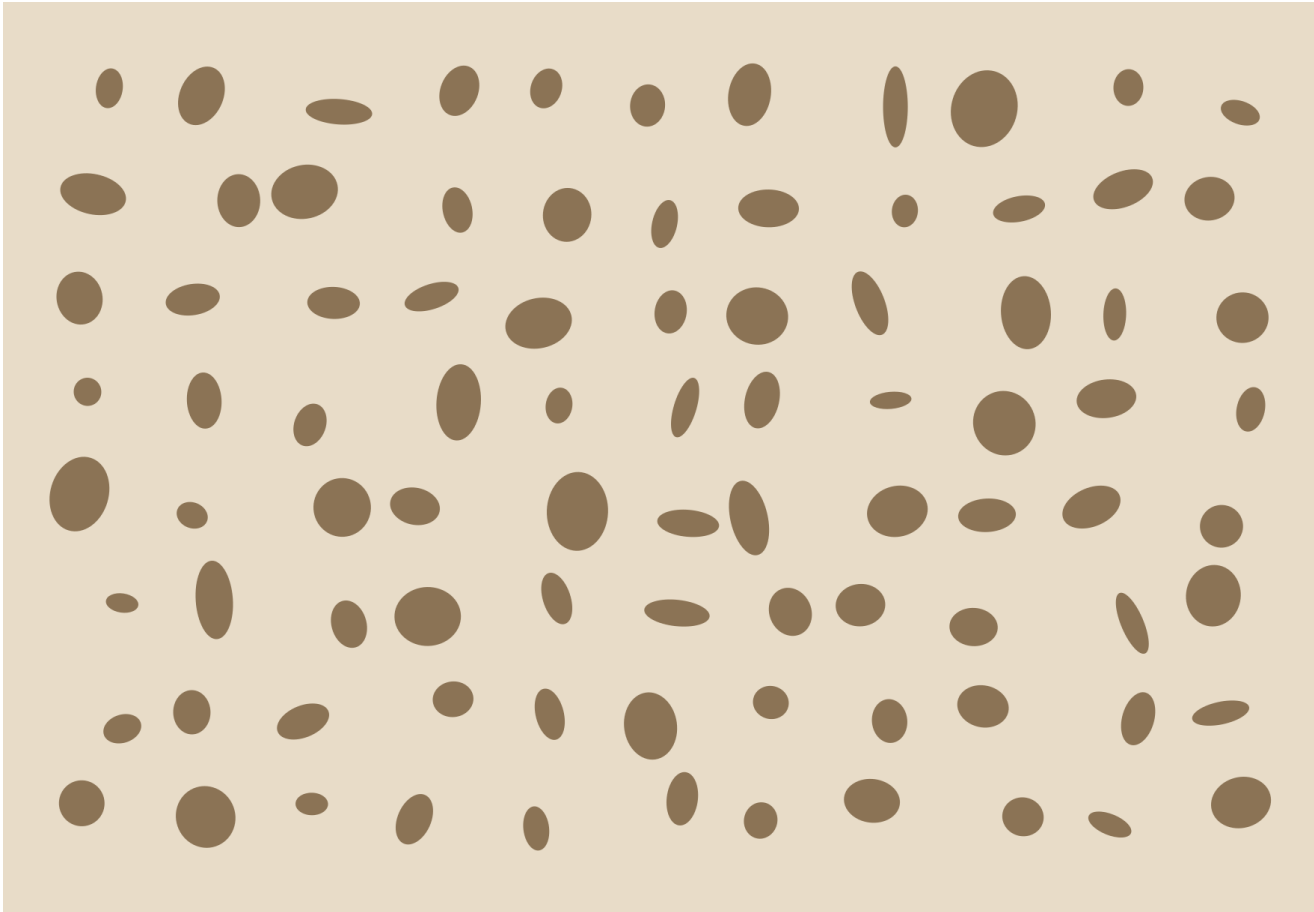
With the **current OCTA Toolbox** capability, we can generate:



A **stimulus** of abstract shapes resembling the pebbles.

- It is ***not* realistic**, but
- **parametrically controlled**

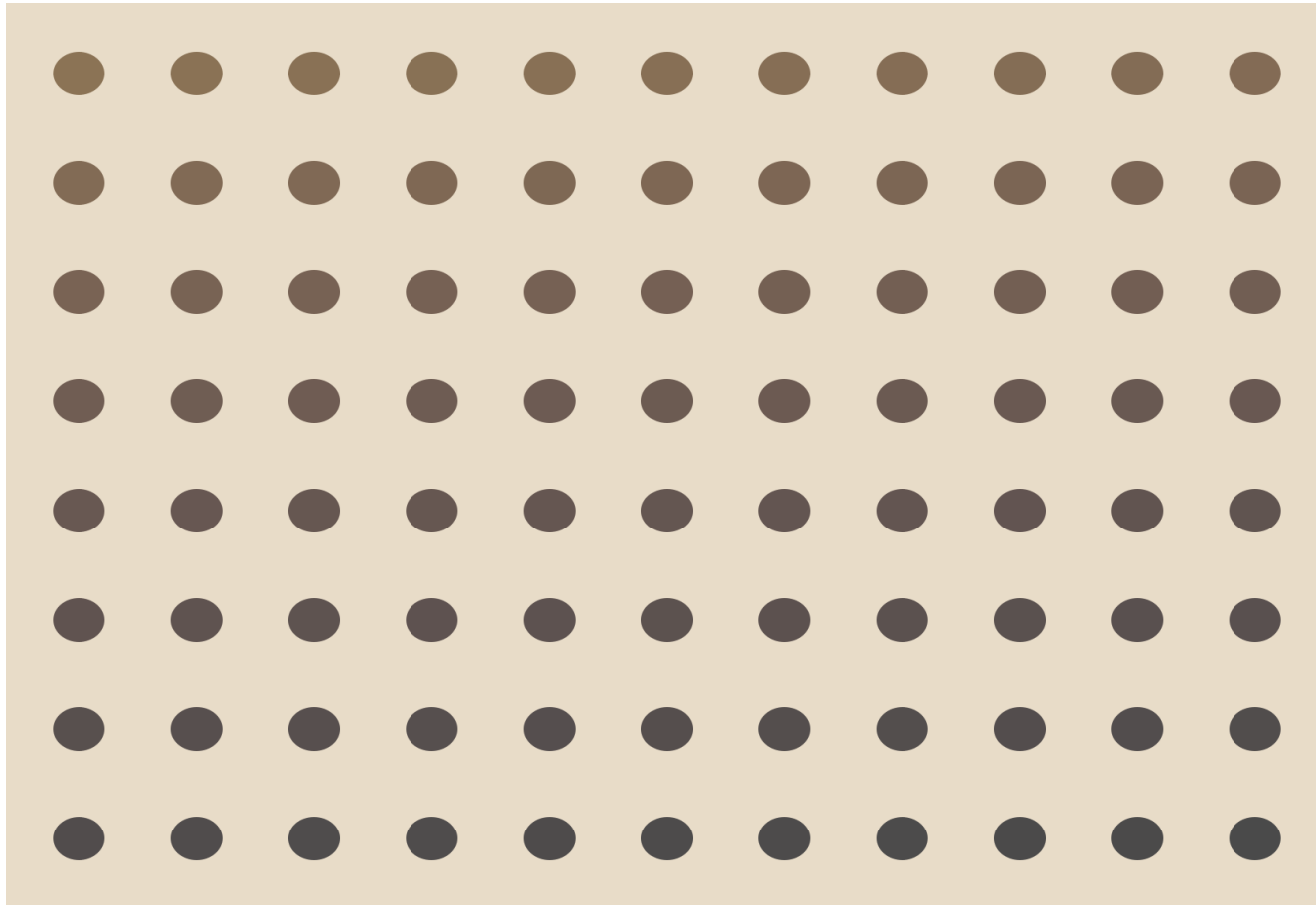
Problem Statement



A **stimulus** of abstract shapes resembling the pebbles.

With modification to:
color of the shapes

Problem Statement



A **stimulus** of abstract shapes resembling the pebbles.

With modification of:
(a lot more) order and
(a lot less) complexity

Problem Statement



Problem Statement

Where this is what we ultimately want:



A **stimulus** of realistic-looking pebbles.

- It should be both **realistic** and
- **parametrically controlled**

Problem Statement



- It should be both **realistic** and
- **parametrically controlled**
(color controlled: dark)

Problem Statement



- It should be both **realistic** and
- **parametrically controlled**
(color controlled: white)

... and ideally a lot more variety of variations among features

Thesis' Goal

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To investigate whether:

preferences for order and complexity in

abstract patterns



translate to



preferences for order and complexity in

realistic-looking images

Thesis' Goal

OCTA toolbox-integrated tool capable of producing stimuli of **realistic-looking vector images**, with **parametric control** capabilities:

To investigate whether:

preferences for order and complexity in

abstract patterns



translate to



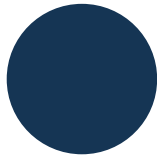
preferences for order and complexity in

realistic-looking images

Thesis' Progress

1
OCTA toolbox-integrated tool capable of producing stimuli of **realistic-looking**
vector images, with **parametric control**
3 4

1.

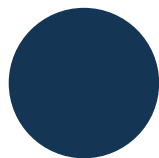


Fully integrated with OCTA

Thesis' Progress

1
OCTA toolbox-integrated tool capable of producing stimuli of **realistic-looking**
vector images, with **parametric control**
3 4

1.



Fully integrated with OCTA

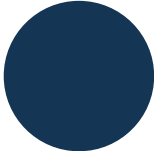
2.



Still work to be done to get the images more recognizable

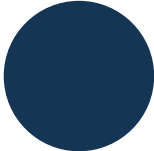
Thesis' Progress

1
OCTA toolbox-integrated tool capable of producing stimuli of **realistic-looking**
vector images, with **parametric control**
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1.  Fully integrated with OCTA
2.  Still work to be done to get the images more recognizable
3.  Generates raster image, not vector *yet*

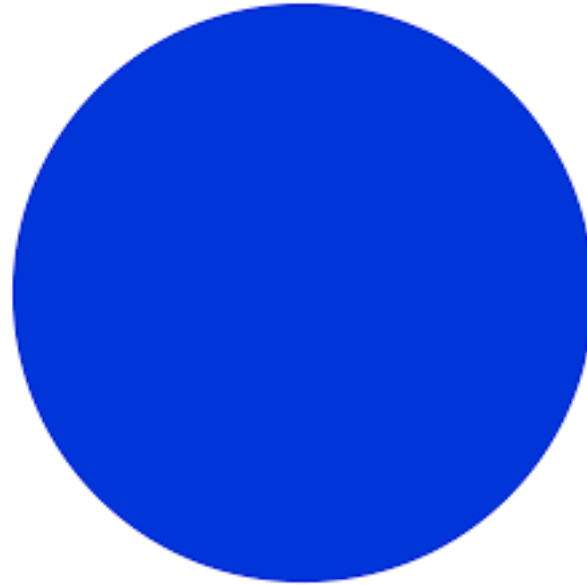
Thesis' Progress

1
OCTA toolbox-integrated tool capable of producing stimuli of **realistic-looking**
vector images, with **parametric control**
3 4

1.  Fully integrated with OCTA
2.  Still work to be done to get the images more recognizable
3.  Generates raster image, not vector *yet*
4.  More variation needed, e.g. color and image style for each individual

Thesis' Progress

Initial results



single realistic raster image generation

Thesis' Progress

Initial results



single realistic raster image generation

Thesis' Progress

Initial results



single realistic raster image generation

Thesis' Progress

Initial results



single realistic raster image generation

Thesis' Progress

Initial results



single realistic raster image generation

Thesis' Progress

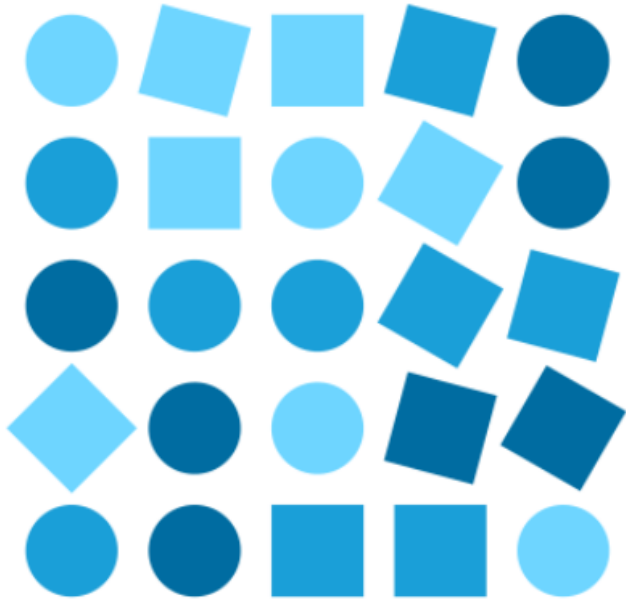
Initial results



single realistic raster image generation

Thesis' Progress

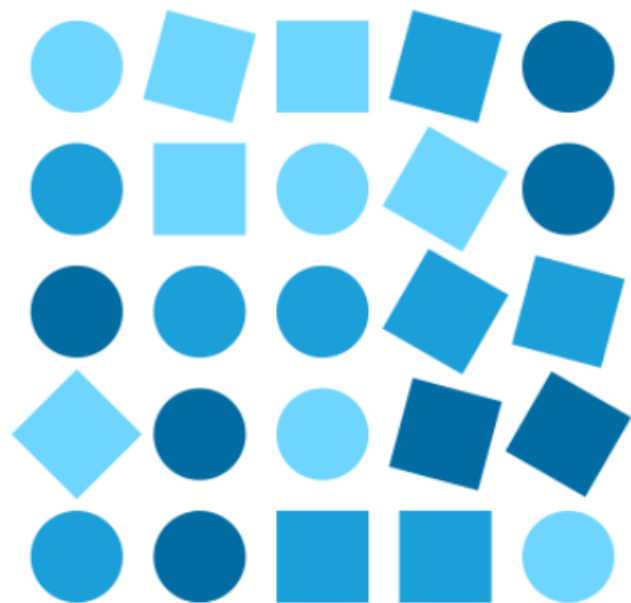
Initial results



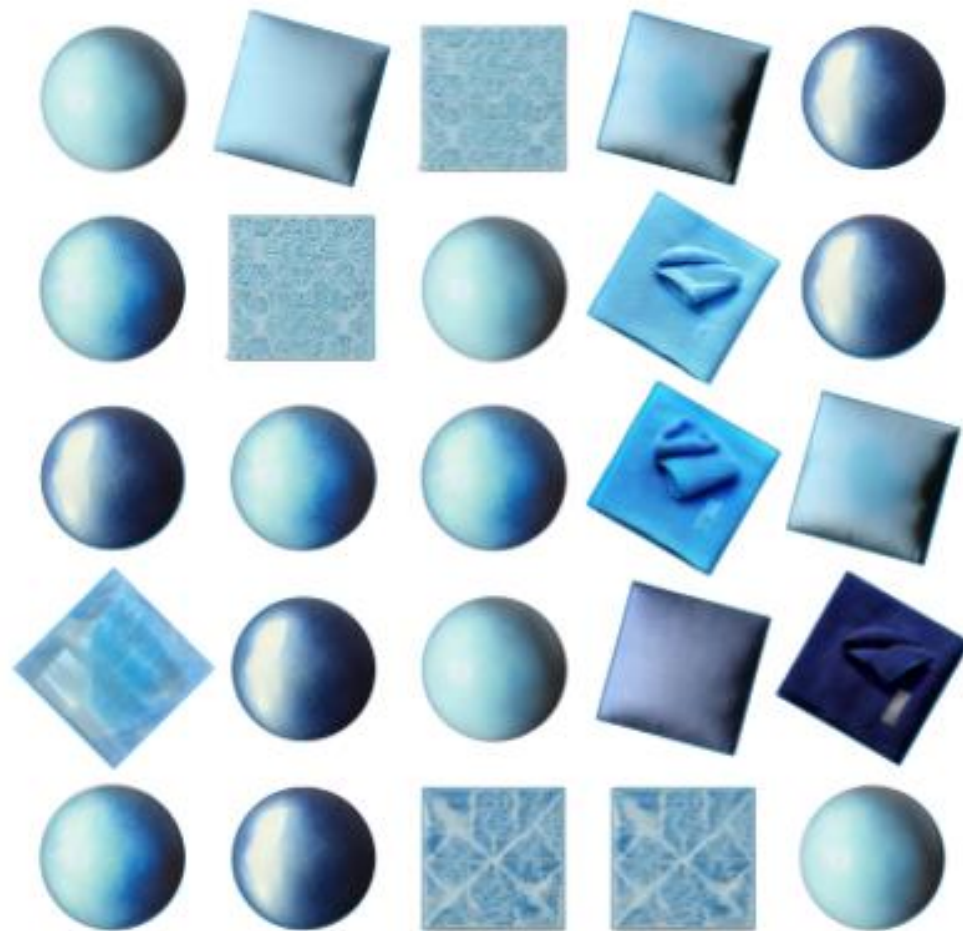
integration with OCTA Toolbox

Thesis' Progress

Initial results



integration with OCTA Toolbox



Methods

1. **OCTA Toolbox**

Generates stimulus consisting of sets of individual shapes

Assigning generated images back to stimulus

2. **ControlNet** with **Stable Diffusion**

Generates images while maintaining input shape

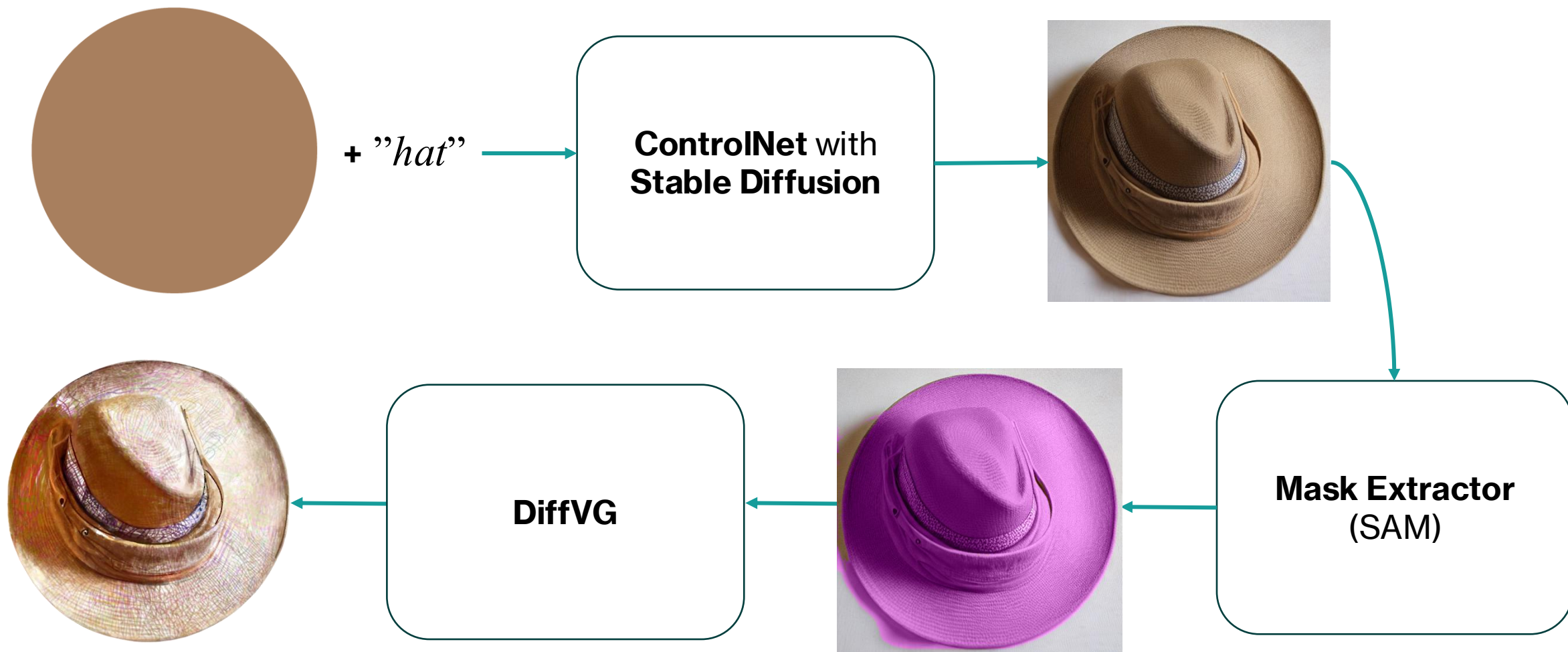
3. **Mask Extractor: SAM**

Decomposes main object and background image

4. **DiffVG**

Vectorizes raster image

Methods



Literature Review

Why **Diffusion**

Stable **Diffusion** is the chosen model to generate the image. Why?

1. Image quality and training stability

Among generative image models:

VAE

- blurry output
- + easy to train

GAN:

- + sharp output
- unstable to train

Diffusion:

- + sharp output
- + stable training
- compute-heavy

Why **Diffusion**

1. Image quality and training stability

VAE

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Diffusion:

- + sharp output
- + stable training
- compute-heavy

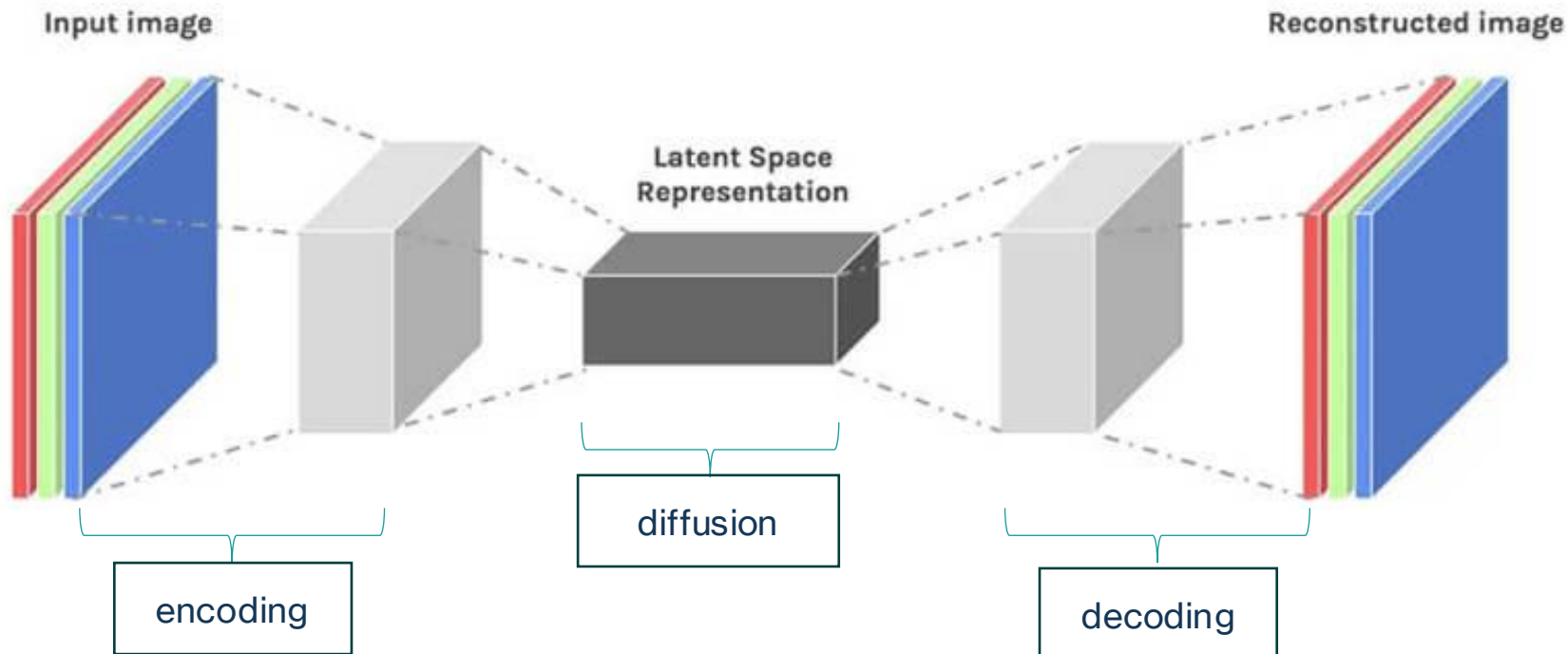


Why **Stable Diffusion**

2. Fast

Stable Diffusion solves the compute problem on **Diffusion**:

by **compressing images** into lower-dimensional latent space (embeddings) before running the diffusion process in that space (Rombach et al., 2022)



Why **Stable Diffusion**

3. Open-source

Diffusion is the standard approach across major image generators, including Midjourney, DALL-E (OpenAI), and Stable Diffusion.

Stable Diffusion is chosen particularly for its **open-source** nature, enabling local running, custom extensions (e.g. ControlNet), and reproducible research.

Literature Review

Why ControlNet

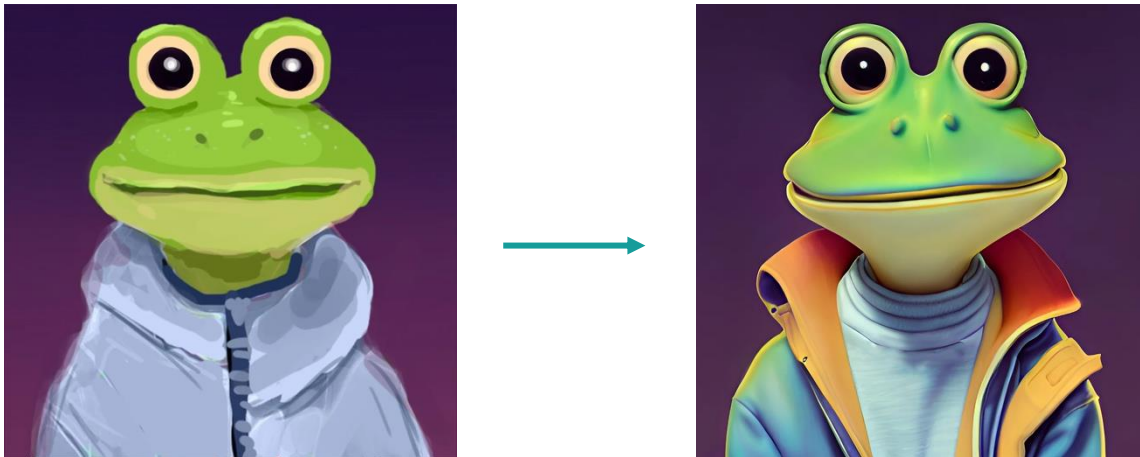


Literature Review

Why ControlNet

Between the methods that can follow abstract shape edges:

- **Img2Img:**
copying entire image (color, texture, background, style),
cannot follow just the edge (Rombach et al., 2022)



Why ControlNet

- **ControlNet:**
extracts **precise edge** information as a dedicated structural guide,
full creative freedom on everything else (style, color, and content) (Zhang et al., 2023)



ControlNet proves to be superior for generating images to follow abstract shapes

Literature Review

Decomposing background images



Background images are an **inevitable byproduct** of diffusion-based image generation process.

Decomposing background images: raster



For **raster images**, background removal process is straightforward:

pixel-based segmentation tools (e.g. Segment Anything Model) easily isolates subject and replaces background with transparency (Kirillov et al., 2023)

Decomposing background images: vector (with mask extractor)

For **vector images**:

a segmentation mask is required to identify foreground / background. Either via:

- **SIVE**: cross-attention maps from a diffusion model (Xing et al., 2024)
- **SAM**: pixel-based segmentation models (Kirillov et al., 2023).

This mask **guides a differentiable renderer** (e.g. DiffVG) to optimize vector within the selected (area of the object) region.

We are planning to use **SAM** for its practicality.



Literature Review

DiffVG

As the last step of the pipeline: **Obtain vector form** from raster image, by having **DiffVG** iteratively optimizes random vector paths until they match (Li et al., 2020)



It is the most common differentiable rasterizer among vector graphics methods, incl.: **SVGDreamer** (Xing et al., 2024), **VectorFusion** (Jain et al., 2023), and **DiffSketcher** (Xing et al., 2023).

Related works

Text -> SVG image

Vector fusion (Jain et al., 2022):

- + One of the first text-to-SVG
- images are **not sharp**
- **no clear object** / background



a train*



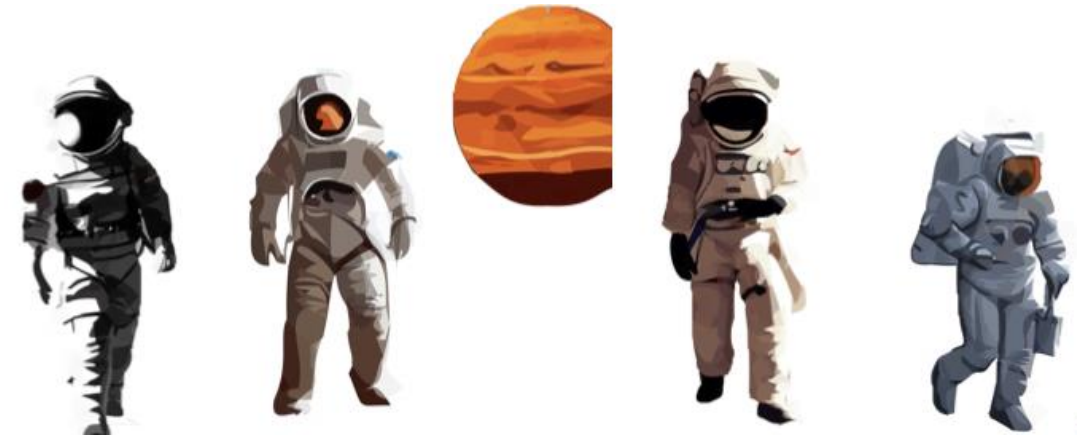
an owl standing on a wire*



A photo of a Ming Dynasty vase on a leather topped table.*

SVGDreamer (Xing et al., 2023):

- + objects are sharp and realistic
- extremely **computationally expensive**

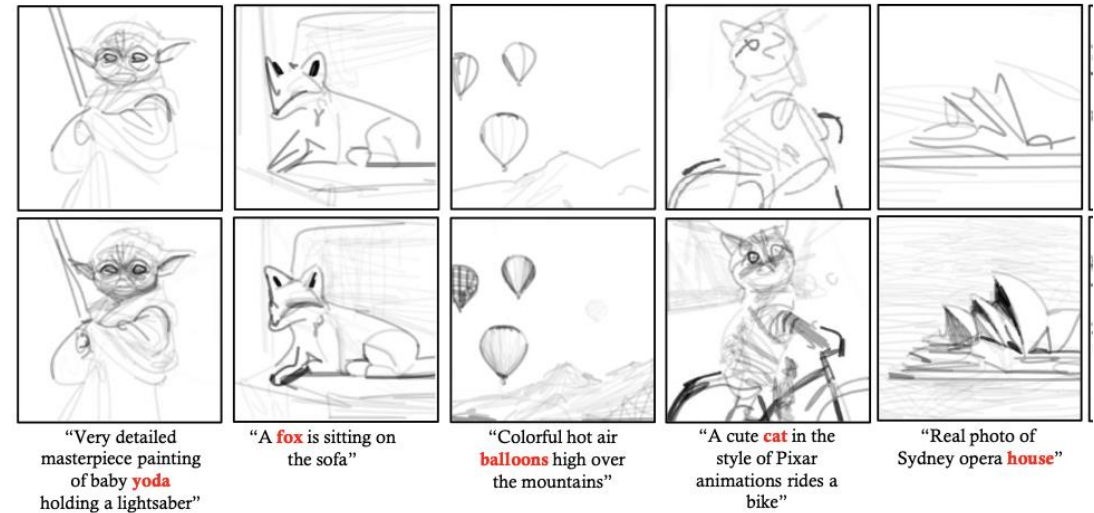


Related works

Raster image + text -> SVG image

DiffSketcher (Xing et al., 2023)

Follows **sketch**, output a more detailed sketch



Vector Painter (Hu et al., 2025):

Follows **only style** from reference, realistic painting-like image



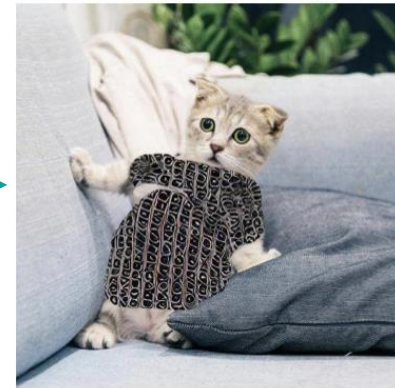
Related works: Shape / edge -> image

Inpainting (Rombach et al., 2022):

Filling in selected area based on the situation around the area of interest.



+ *"A black formal dress on the kitten"* →



+ *"A purple scarf around the neck"* →

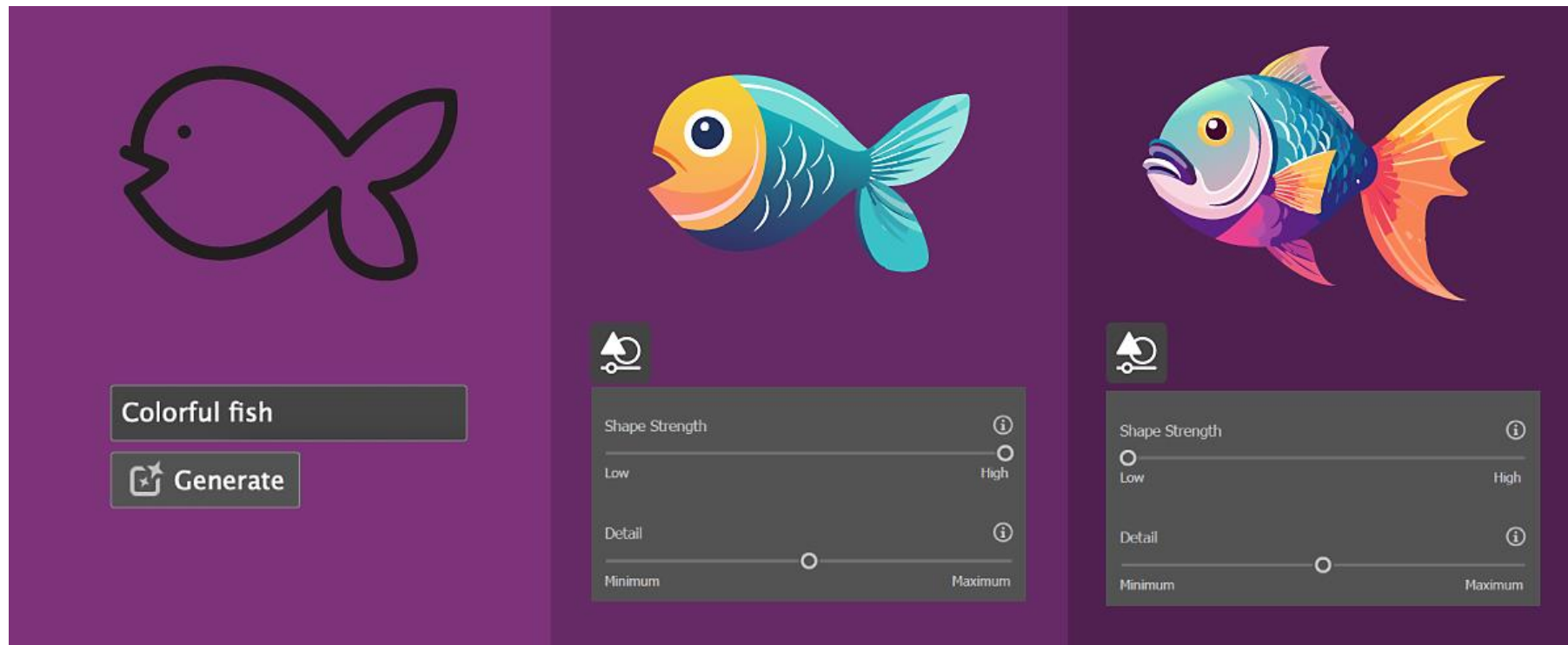


Related works: Shape / edge -> image

Adobe's Generative Shape Fill (Adobe, n.d.) :

The closest thing to **shape to realistic-looking SVG** implementation.

Closed-source, no publications.



Current limitations + plans

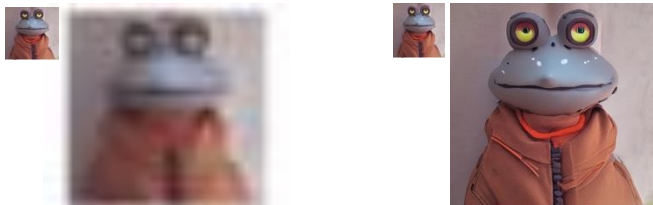
- **Recognizability**



- **Color accuracy**



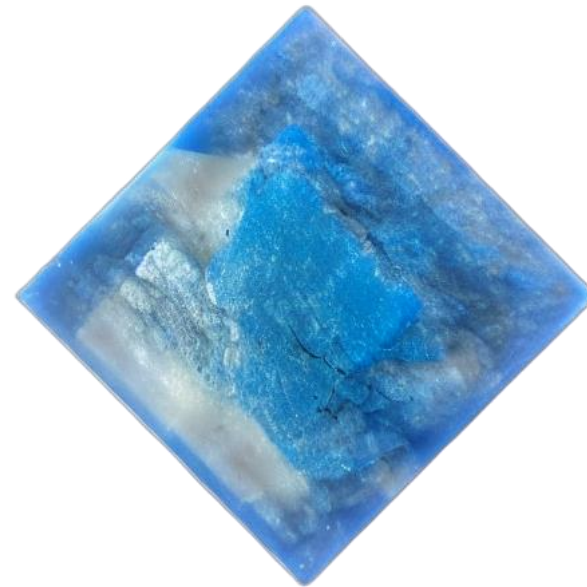
- **Raster format**



Recognizability

Realism and recognizability are expected for our purposes.

While these images are ***sufficiently realistic***, they are **not easily recognizable** as: '*tile*', '*bowling ball*', and '*blue crystal*'.



Potential fix: better prompt choice (e.g. '*pizza*' in place of '*bowling ball*')

Color accuracy

Generated images should **carry over the color information** (as it is one of the feature/dimension provided in OCTA Toolbox). This has not been implemented.



Potential fix: post-processing with CV techniques

Raster vs vector format

Vector image is **preferred** over raster format for the thesis:

It is **scalable** and **suits OCTA Toolbox's** primary design.



Potential fix: apply DiffVG

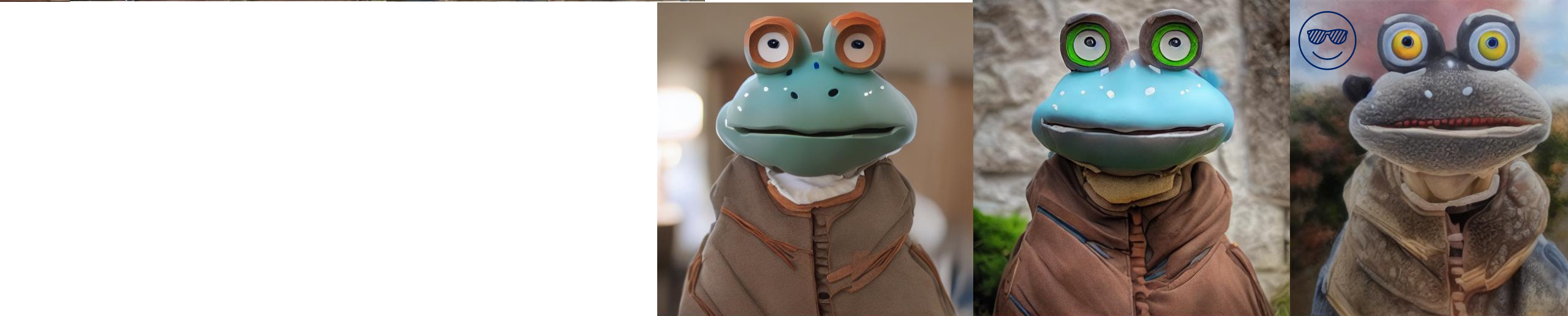
End of slide

Thank you for the time and attention



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References

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- Zhang, L., Rao, A., & Agrawala, M. (2023). Adding conditional control to text-to-image diffusion models. *arXiv preprint arXiv:2302.05543*. <https://doi.org/10.48550/arXiv.2302.05543>

Modifications on slide since last update (Thu 19 Feb)

- Add the real world image -> OCTA generated abstract stimulus -> OCTA + AI generated realistic looking stimulus.
- Add inpainting as (one of the) related work(s).
- Removal of StarVector from related works
- Add output sample for image generator model (VAE vs GAN vs diffusion) comparison
- Add GIF showcasing parametric control of OCTA Toolbox (from its Shiny app)*
- A new version of the comparison of abstract stimulus to realistic looking stimulus
- Addition on removal bg: “We use SAM”
- Addition of Current limitations:
 - Color accuracy
 - Recognizability in some objects
 - Raster v vector

Next steps for technical

- Make the objects more recognizable:
 - Playing with the prompts
 - The negative prompt of "background"
- 2d vs 3d
- Color control:
 - Straight from the model
 - Post modification
 - Parameters for how many colors (helps with realistic images)x
- Contour optimization algorithm (add a bit of shadow) to help make it recognizable
- Edge optimization
 - use gaussian blur to improve the generation
- Add limitations:
 - Vectors
 - Color
 - Edge accuracy
 - Recognizability